

1356. Sort Integers by The Number of 1 Bits

Difficulty: Easy

Link: <https://leetcode.com/problems/sort-integers-by-the-number-of-1-bits/description/?envType=daily-question&envId=2026-02-25>

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Problem:

You are given an integer array `arr`. Sort the integers in the array in ascending order by the number of `1`'s in their binary representation and in case of two or more integers have the same number of `1`'s you have to sort them in ascending order.

Return *the array after sorting it*.

Example 1:

Input: `arr = [0,1,2,3,4,5,6,7,8]`

Output: `[0,1,2,4,8,3,5,6,7]`

Explanation: `[0]` is the only integer with 0 bits.

`[1,2,4,8]` all have 1 bit.

`[3,5,6]` have 2 bits.

`[7]` has 3 bits.

The sorted array by bits is `[0,1,2,4,8,3,5,6,7]`

Example 2:

Input: `arr = [1024,512,256,128,64,32,16,8,4,2,1]`

Output: `[1,2,4,8,16,32,64,128,256,512,1024]`

Explantion: All integers have 1 bit in the binary representation, you should just sort them in ascending order.

Constraints:

- `1 <= arr.length <= 500`
- `0 <= arr[i] <= 104`